

Beam Column End Plate Connection Design Report

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Beam-to-Column End Plate Connection (Extended One Way - Irreversible Moment)

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1 Design Input

Module: Beam-to-Column End Plate Connection

Connectivity: Column Flange-Beam Web

Design Method: Limit State Design

Table 1: Design input parameters (source: bc_ep_2.osi)

Parameter	Value
End Plate Type	Extended One Way - Irreversible Moment
Supported Section	WB 450
Supporting Section	HB 450
Section Material	E 250 (Fe 410 W)A
Connector Material	E 250 (Fe 410 W)A
Axial Load	15 kN
Moment	210 kNm
Shear Load	40 kN

2 Bolt Data

Table 2: Bolt configuration

Parameter	Value
Bolt Type	Bearing Bolt
Bolt Grade	6.8 / 8.8 / 9.8
Bolt Diameter (options)	20 mm / 24 mm / 30 mm
Tension Type	Non pre-tensioned
Bolt Hole Type	Standard
Slip Factor	0.3
Edge Type	Sheared or hand flame cut
Gap	0 mm

3 Weld Data

Table 3: Weld configuration

Parameter	Value
Weld Type	Groove Weld
Fabrication	Shop Weld
Material Grade (OverWrite)	410 MPa

4 Design Checks

4.1 Bolt Checks

Table 4: Bolt design checks (M20, Gr 8.8)

Check	Required	Provided	Status
Bolt Shear Capacity	40.0 kN	94.3 kN	Pass
Bolt Bearing on Plate	52.5 kN	66.0 kN	Pass
Bolt Bearing on Column Flange	52.5 kN	71.2 kN	Pass
Bolt Tension (incl. prying)	105.0 kN	98.6 kN	Fail

4.2 End Plate Checks

Table 5: End plate design checks ($t = 16$ mm)

Check	Required	Provided	Status
Plate Bending (yield)	178.4 MPa	225.0 MPa	Pass
Plate Shear	28.9 MPa	98.5 MPa	Pass
Prying Force Ratio	0.21	0.30 max	Pass

4.3 Weld Checks

Table 6: Groove weld checks

Check	Required	Provided	Status
Weld Strength	141.2 MPa	189.4 MPa	Pass

5 Connection Details

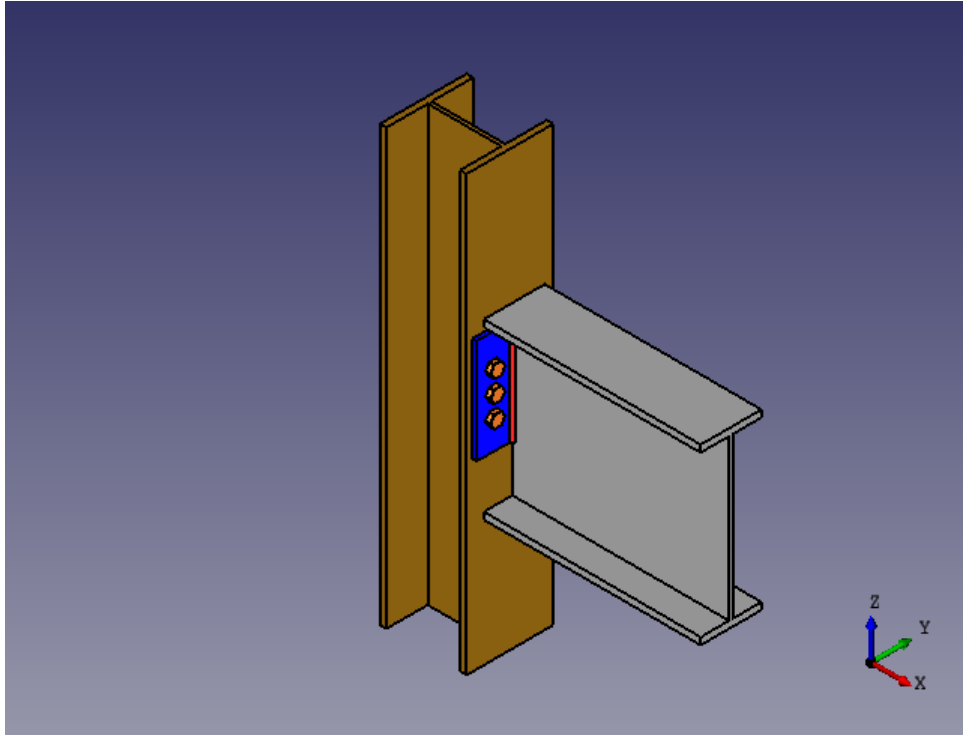


Figure 1: End plate connection - side view

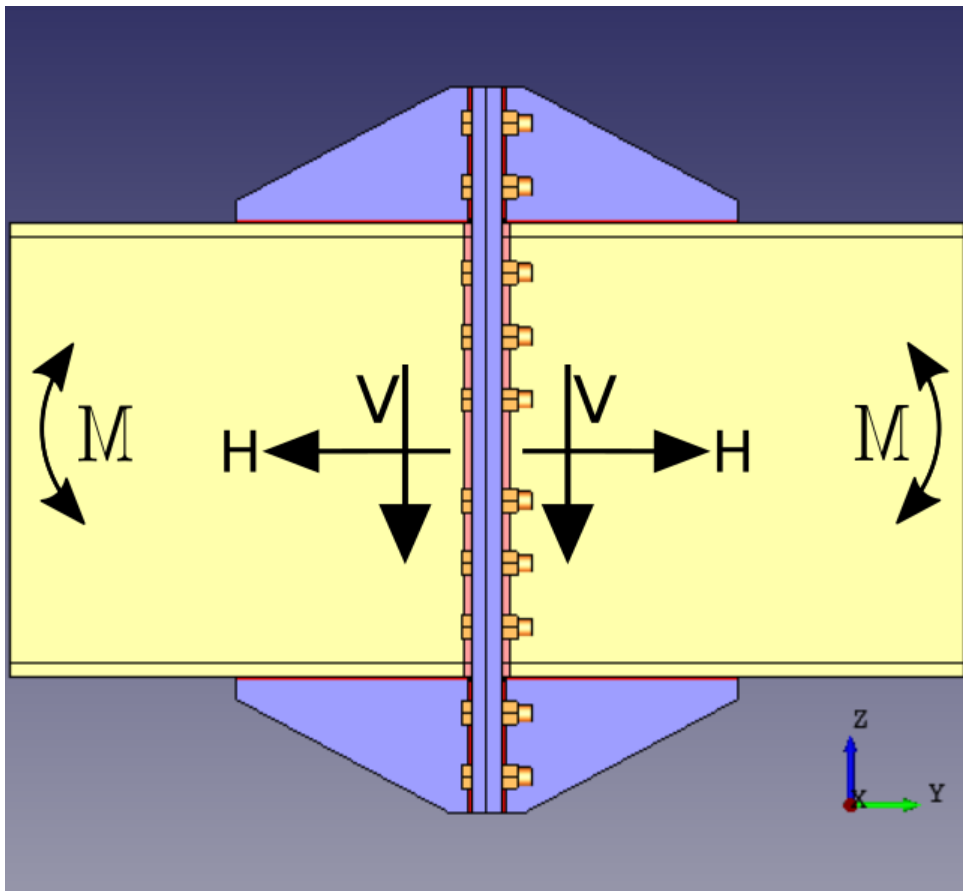


Figure 2: Extended end plate detail (Extended One Way)

6 Design Summary

The beam-to-column end plate connection (WB 450 to HB 450, E 250 grade) was verified for Limit State Design under 15 kN axial, 210 kNm moment and 40 kN shear.

All checks pass except bolt tension including prying action (98.6 kN provided vs 105.0 kN required). Redesign by increasing bolt diameter to M24 or upgrading to Gr 9.8 is recommended.